MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPES SHOF, DHOF, THOF, AND FHOF

(F) This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the cable described herein shall consist of this document and the latest issue of Specification MIL-C-915.

REQUIREMENTS:
Qualification required, except for Type SHOF.

Construction

First - Copper conductor, uncoated, (see Details for size).
Second - Separator.
Third - Synthetic rubber insulation, (see Details for wall thickness). Standard Identification Code, applied by method 3 or 4.
NOTE: A colored rubber filled cotton tape may be used on size 23 and larger.
Fourth - Reinforcement on type SHOF, sizes 23 and larger.
Fifth - The required number of conductors cabled together with a left-hand lay (see Details for length of lay). Fibrous fillers shall be employed on multi-conductor types to obtain a firm, well-rounded assembly.
Sixth - Binder tape applied helically with overlap on multi-conductor cables.
Seventh - Jacket of polychloroprene or standard thermoplastic on sizes 42 and smaller of all types, except on THOF-14. Jacket of polychloroprene on THOF-14 and on sizes 60 and larger of all types. All jackets shall be black. Cable surface marking required.

NOTE: The initial letter of the cable designation denotes the number of conductors. Thus "S" denotes a single-conductor cable; "D" a two-conductor cable; "T" a three-conductor cable; and "F" a four-conductor cable.

Details

Type and size	Condr. size Navy stand.	Insulation thick. nom. (inch)	Lay of condr. max (inches)	Overall diameter min (inches)	Overall diameter max (inches)	Roller dia. for bend. endurance max (inches)	Condr. res. per 1000 ft max (ohms)	Insulation resistance (megohms)
SHOF-3	2-1/2(65)	0.031		0.195	0.210		4.26	300
SHOF-23	23(228)	.040		.440	.460		.499	300
SHOF-60	60(304)	.050		.570	.600		.187	250
SHOF-150	150(760)	.070		.830	.870		.0747	200
SHOF-200	200(988)	.070		.940	.980		.0575	200
SHOF-250	250(1254)	.070		1.035	1.085		.0453	150
SHOF-500	500(259)	.090		1.380	1.450		.0234	100
SHOF-650	650(427)	.100		1.540	1.610		.0154	100
SHOF-800	800(4033)	.100		1.600	1.670		.0141	100

(F) Denotes changes

FSC 6145

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Details (con)

Type and size	Condr. size Navy stand.	Insulation thick. nom. (inch)	Lay of condr. max (inches)	Overall diameter min (inches)	Overall diameter max (inches)	Roller dia. for bend. endurance max (inches)	Condr. res. per 1000 ft max (ohms)	Insulation resistance (megohms)
DHOF-3	2-1/2(26)	.031	2	.405	.425	1.3	4.31	300
DHOF-4	4(41)	.031	2	.440	.460	1.4	2.73	250
DHOF-6	6(65)	.031	2-1/2	.490	.510	1.5	1.72	200
DHOF-9	9(90)	.031	3	.540	.570	1.7	1.24	200
DHOF-14	14(140)	.040	4	.675	.705	2.1	.802	200
DHOF-23	23(228)	0.040	5-1/2	0.820	0.860	3.0	0.509	150
DHOF-30	30(304)	.050	7-1/2	.920	.960	3.5	.381	150
DHOF-83	83(418)	.070	16	1.390	1.450	7.0	.139	150
DHOF-250	250(1254)	.070	24	2.000	2.100		.0462	100
DHOF-400	400(2052)	.090	30	2.400	2.500		.0283	100
THOF-3	2-1/2(26)	.031	2	.430	.450	1.4	4.31	300
THOF-4	4(41)	.031	2-1/2	.460	.480	1.4	2.73	250
THOF-6	6(65)	.031	3	.520	.550	1.7	1.72	200
THOF-9	9(90)	.031	3-1/2	.570	.600	1.8	1.24	200
THOF-14	14(140)	.040	4-1/2	.720	.750	2.3	.802	200
THOF-23	23(228)	.040	6	.860	.900	3.5	.509	150
THOF-42	42(209)	.070	12	1.200	1.250	4.5	.277	150
THOF-150	150(760)	.070	18	1.740	1.820		.0762	125
THOF-250	250(1254)	.070	24	2.140	2.240		.0462	100
(F) THOF-400	400(2052)	.090	30	2.680	2.800		.0283	100
(F) THOF-500	500(5054)	.090	33	2.920	3.100	---	.0230	100
THOF-600	600(427)	.100	36	2.980	3.150		.0200	100
FHOF-3	2-1/2(26)	.031	2-1/2	.460	.480	1.4	4.31	300
FHOF-4	4(41)	.031	3	.520	.550	1.6	2.73	250
FHOF-9	9(90)	.031	4	.630	.660	1.8	1.24	200
FHOF-42	42(209)	.070	15	1.300	1.380	4.1	.277	150
FHOF-60	60(304)	.070	18	1.430	1.510	4.5	.191	150
FHOF-133	133(684)	.070	20	1.920	2.000		.0847	125

(F) NOTE 1: For conductor size Navy Standard 500 (5054), nominal strand diameter = 0.010-inch; minimum number of strands = 5054; nominal conductor diameter = 0.960-inch; conductor cross-sectional area = 505,400 circular-mils nominal; 495,300 circular-mils minimum.

EXAMINATION AND TESTS:

Requirements:

(F) Basic Electrical:

(F) Conductor resistance - Ohms/1000 ft. at 25° Celsius (C), maximum . . (See Details)

Voltage withstand - Volts, root mean square (rms), minimum

sizes 3 to 9 inclusive 2000

sizes 14 to 800 inclusive. 2500

Insulation resistance - Megohms/1000 ft., minimum. . . . . (See Details)

Group A:

Visual and dimensional

Group B:

Cold bending, cable - (For type SHOF only) at minus 30° + 2°C. Two turns around mandrel with diameter eight times that of specimen . . . . . (No damage)

Physicals (unaged)

Insulation (conductor)

(F) Tensile strength - pounds per square inch gage (lb/in^2g), minimum 600

Elongation - Percent, minimum 300

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	Requirements:
Group B (con):	
Physicals (unaged) (con)	
Jacket (cable)	
Polychloroprene:	
(F) Tensile strength - lb/in^{2}g, minimum	1800
Elongation - Percent, minimum	300
Set - Inch, maximum	3/8
Standard thermoplastic:	
Tensile strength - lb/in^{2}g, minimum	1800
(F) Elongation - Percent, minimum	125
Twisting and bending endurance - (Note 1)	
At 75°C., cycles, minimum	6000
At minus 30°C., cycles, minimum	2000
(F) Note 1: Tests to be performed on all sizes of DHOF, THOF, and FHOF, excepting DHOF-250, DHOF-400, THOF-150, THOF-250, THOF-400, THOF-500, THOF-600, FHOF-133. Tests not required for SHOF. (See details for roller sizes).	
Group C:	
Flammability - Inches, maximum	2
Physicals (aged)	
Insulation (conductor)	
Air pressure heat	
Tensile strength - Percent of unaged, minimum	70
Elongation - Percent of unaged minimum.	75
Jacket (cable) (Polychloroprene)	
Air pressure heat	
Tensile strength - Percent of unaged, minimum.	50
Elongation - Percent of unaged, minimum.	50
Hot oil immersion	
Tensile strength - Percent of unaged, minimum	60
Elongation - Percent of unaged, minimum.	60
Jacket (cable) (Standard thermoplastic)	
Hot oil immersion	
Tensile strength - minimum	1400
Elongation - Percent - Minimum	50
Permanence of printing (jacket) - Cycles, minimum	250

QUALIFICATION INSPECTION

Qualification inspection shall include basic electrical plus all of Groups A, B and C.

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
SHOF	- 23	2000
SHOF	- 60 thru 250.	1000
SHOF	- 500 thru 800.	500
DHOF	- 6	2000
DHOF	- 9 and 14.	1500
DHOF	- 23 thru 83	1000
DHOF	- 250 and 400	500
THOF	- 6	2000
THOF	- 9 and 14.	1500
THOF	- 23 thru 150.	1000
THOF	- 250 thru 600	500
FHOF	- 4	2000
FHOF	- 9	1500
FHOF	- 42 and 60.	1000
FHOF	- 133	500

NOTE 1. (SHOF-3, DHOF-3, DHOF-4, THOF-3, THOF-4, FHOF-3) Packed: 500 feet (nominal) per coil.

Custodians:
Army - MI
Navy - SH
Air Force - 80

Review activities:
Army - EL, MI, AV
(F) Navy - SH, EC
Air Force - 80

User activities:
Army - ME, MU, WC
(F) Navy - CG, YD

Preparing activity:
Navy - SH
(Project 6145-0674-06)

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